Achyut Nair

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Education

University of Texas at Austin — Austin, TX
B.S. in Chemical Engineering, Minor in Economics; GPA: 3.82 — *Expected May 2028*

- Affiliations: Longhorn Racing Combustion, AIChE Officer, Materials Science and Engineers, Energy Institute; Valero Summit Scholar and Dwight E. Huth Presidential Scholar

Experience

Valero Energy Corporation — Ardmore, OK
Environmental Engineering Intern — *May 2026 – August 2026*

- Identified an outdated, manual Title V emissions calculation process and rebuilt it into modular, low-input Excel models, cutting reporting time by 50 hours and standardizing the workflow sitewide
- Evaluated cost assumptions against operational requirements for the Influent Lift Station capital project and authored the technical and economic proposal that guided the investment decision
- Installed a Vapor Combustion Unit (VCU) flowmeter to enable direct process measurement, reducing compliance risk and improving emissions reporting accuracy
- Built real-time Critical Instrumentation monitoring software using Statistical Process Control (SPC) to flag flatline, drift, and out-of-control deviations

Longhorn Racing Combustion — Austin, TX
Composites Engineer — *August 2025 – Present*

- Built Python workflows to automate post-processing of ANSYS Fluent simulation outputs, cutting manual analysis time 85% and standardizing the team's pipeline
- Ran finite element analysis (FEA) on the undertray to design carbon fiber layups that met structural requirements within cost and material constraints
- Authored vacuum-assisted resin infusion documentation, standardizing fabrication steps across the composites team

Röchling Industrial Composites — Cleveland, OH
Research and Development Intern — *May 2025 – August 2025*

- Built ANSYS simulation workflows to supplement experimental testing, saving 105 hours and $25,000 in Q3
- Analyzed experimental process data across 3 pultrusion lines to identify key performance parameters
- Performed FTIR and DSC characterization to evaluate resin chemistry and flow behavior

Energy Institute — Austin, TX
Student Assistant — *August 2024 – Present*

- Built a dedicated website for the Energy Week Conference, improving attendee experience and event communication
- Programmed a digital check-in system that cut system-wide delays by 2 hours daily
- Edited video content in Adobe Premiere that increased online visibility by 47%

Projects

Longhorn Cube | Research Member — January 2025

- Reviewed literature on CO2 adsorbents and identified biochar as a cost-effective, scalable capture solution
- Modeled capture scenarios in Excel to minimize levelized cost of capture (LCOC) across sorbent efficiencies, plant scale, and throughput

Technical Skills

Programming & Computing: Python (NumPy, pandas, scikit-learn), Bash, Linux, MATLAB, HTML/CSS/JS
Process & Compliance: SPC, Six Sigma, Title V Emissions Compliance
Engineering & Analysis: ANSYS, DFT (VASP), SOLIDWORKS, Finite Element Analysis (FEA)
Materials & Characterization: FTIR, DSC, Carbon Fiber Composites, Fiberglass Composites
Digital & Content Tools: Adobe Premiere, HTML/CSS/JS, Web Development

Awards & Interests

Achievements: National Speech and Debate Academic All American, 2X State Runner-up in Biotechnology Design, NASA Highschool Aerospace Scholars
Interests: Baking, Weight Lifting, Hiking, Podcasting (@Donesimply)